Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the *C. elegans* Box C/D complex - Supplementary Information

Micha J. Birklbauer[1,2*], Fränze Müller[3], Sowmya Sivakumar Geetha[4,5,6], Manuel Matzinger[3], Karl Mechtler[3,7,8], Viktoria Dorfer[1*]

[1*]Bioinformatics Research Group, University of Applied Sciences Upper Austria, Softwarepark 11, Hagenberg, 4232, Upper Austria, Austria.
[2*]Institute for Symbolic Artificial Intelligence, Johannes Kepler University Linz, Altenberger Straße 69, Linz, 4040, Upper Austria, Austria.
[3]Institute of Molecular Pathology (IMP), Vienna BioCenter (VBC), Campus-Vienna-Biocenter 1, Vienna, 1030, Vienna, Austria.
[4]Max Perutz Labs (MPL), Vienna BioCenter (VBC), Dr. Bohr-Gasse 9/ Vienna Biocenter 5, Vienna, 1030, Vienna, Austria.
[5]Max Perutz Labs (MPL), Department of Chromosome Biology, University of Vienna, Dr. Bohr-Gasse 9/ Vienna Biocenter 5, Vienna, 1030, Vienna, Austria.
[6]Vienna BioCenter PhD Program, a Doctoral School of the University of Vienna and the Medical University of Vienna, Vienna BioCenter (VBC), Dr. Bohr-Gasse 9/ Vienna Biocenter 5, Vienna, 1030, Vienna, Austria.
[7]Institute of Molecular Biotechnology (IMBA), Austrian Academy of Sciences, Vienna BioCenter (VBC), Dr. Bohr-Gasse 3, Vienna, 1030, Vienna, Austria.
[8]Gregor Mendel Institute (GMI), Austrian Academy of Sciences, Vienna BioCenter (VBC), Dr. Bohr-Gasse 3, Vienna, 1030, Vienna, Austria.

*Corresponding author(s). E-mail(s): micha.birklbauer@fh-hagenberg.at; viktoria.dorfer@fh-hagenberg.at;

Contributing authors: fraenze.mueller@imp.ac.at;
sowmya.sivakumar.geetha@univie.ac.at; manuel.matzinger@imp.ac.at;
karl.mechtler@imp.ac.at;

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1 Encoding of mass spectra and peptides

In order to demonstrate the process of encoding mass spectra and peptides as vectors, consider the following:

1. Peptide P which is a float array of calculated m/z values for all theoretical ions.
2. Integer number p of theoretical ions in P.
3. Mass spectrum S which is a float array of m/z values of its peaks. (The intensities are not used for this algorithm).
4. Integer number s of peaks in S.
5. Tolerance T which is single float value and is given in Dalton.

The mass spectrum encoding vector is calculated as follows (here implemented in the C++ language):

```
const int dims = 500000;
const double n = 0.39894228040143267793994605993438;
float normpdf(float x, float mu, float sigma) {
    if (sigma == 0.0) {
        return 1.0;
    }
    return (n/sigma) * exp(-0.5*squared((x-mu)/sigma));
}
float t = round(T * 100.0f);
auto* v = new float[dims] {0.0}; \\ the encoding vector
for (int i = 0; i < s; ++i){
    int currentPeak = (int) round(S[s] * 100.0f);
    int minPeak = currentPeak - t > 0 ?
        currentPeak - t :
        0;
    int maxPeak = currentPeak + t < dims ?
        currentPeak + t :
        dims - 1;
    for (int j = minPeak; j <= maxPeak; ++j) {
        float currentVal = v[j];
        float newVal = normpdf((float) j,
                               (float) currentPeak,
                               t / 3.0f);
        v[j] = std::max(currentVal, newVal);
    }
}
```

The peptide encoding vector is calculated as follows (here implemented in the C++ language):

```
const int dims = 500000;
auto* v = new float[dims] {0.0}; \\ the encoding vector
int index = 0;
for (int i = 0; i < p; ++i){
    index = (int) round(P[p] * 100.0f);
    v[index] = 1.0f;
}
```

Furthermore, a graphical explanation of the mass spectrum encoding is given in Supplementary Figure 8 and the peptide encoding is shown in Supplementary Figure 9. The complete source code for encoding and matrix multiplication is given in https://github.com/hgb-bin-proteomics/CandidateSearch and https://github.com/hgb-bin-proteomics/CandidateVectorSearch.

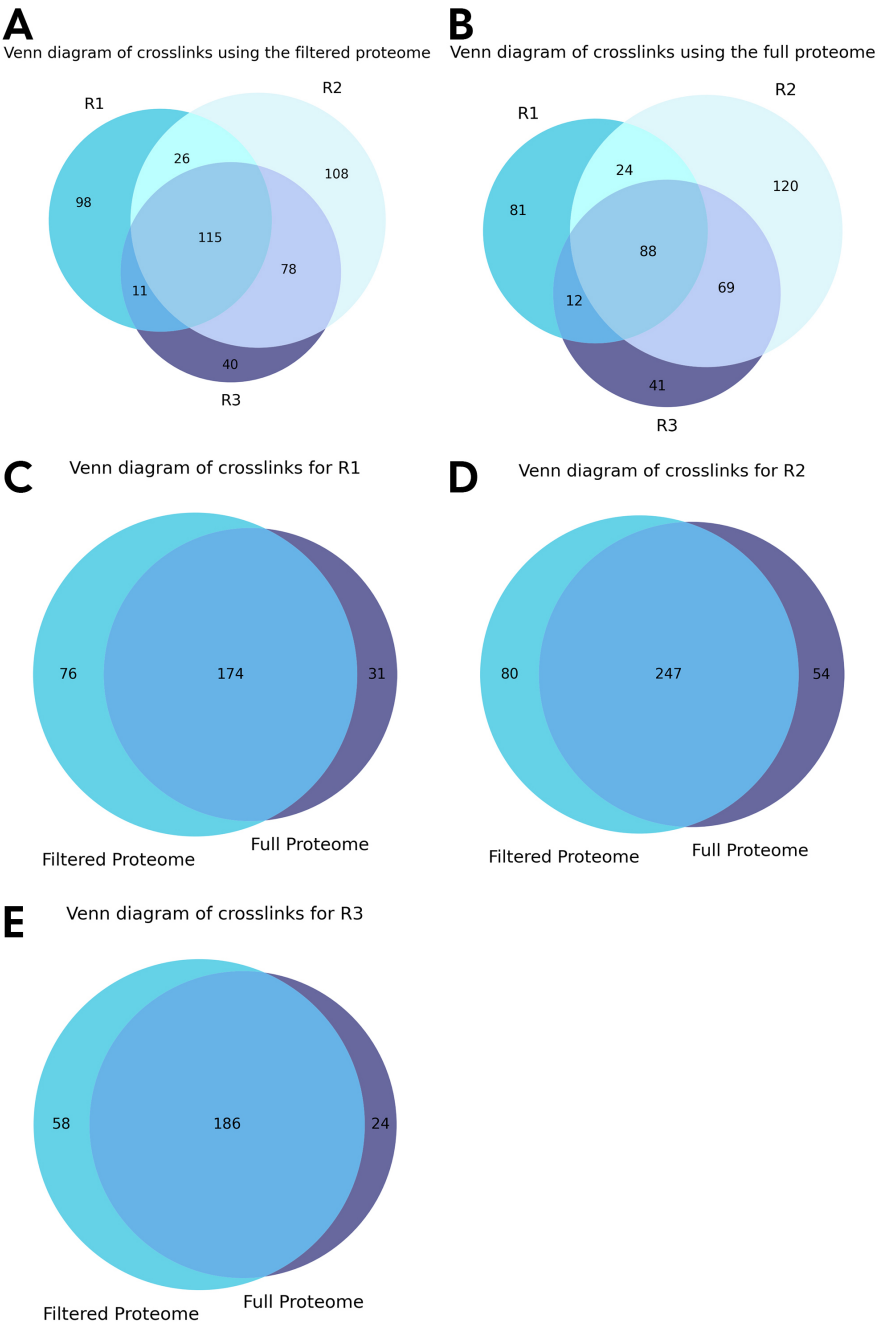


Fig. 1 Overlaps of identified crosslinks at 1% FDR in the *C. elegans* dataset using MS Annika for search and xiFDR [1] for validation. A) Overlaps of identified crosslinks of the three biological replicates using the filtered proteome. B) Overlaps of identified crosslinks of the three biological replicates using the full proteome. C) Overlap of identified crosslinks for filtered and full proteome in replicate 1. D) Overlap of identified crosslinks for filtered and full proteome in replicate 2. E) Overlap of identified crosslinks for filtered and full proteome in replicate 3.

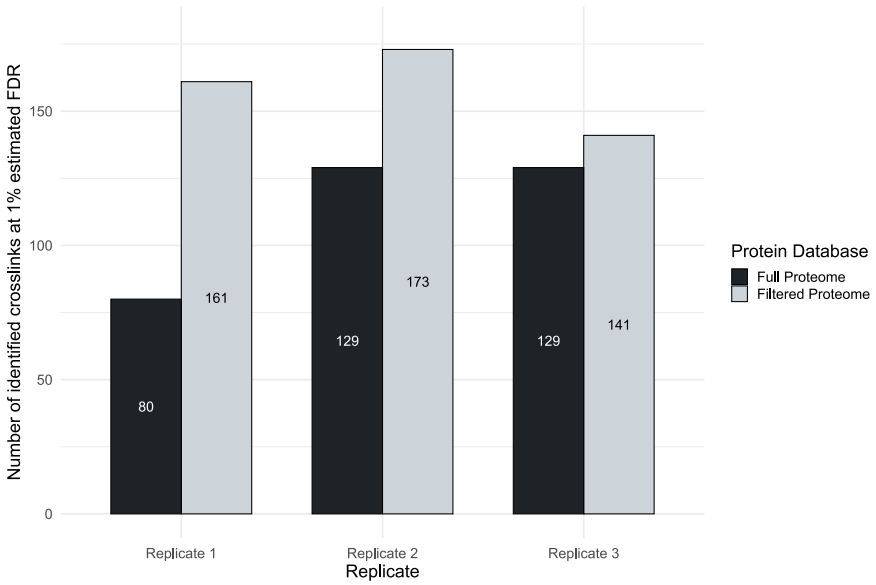


Fig. 2 Influence of protein database size on the non-cleavable search of MS Annika when using the built-in valdiation algorithm. Mass spectrometry data of *C. elegans* nuclei was searched in MS Annika once using the full *C. Elegans* proteome (n = 26 695) and once using a filtered proteome of abundant proteins identified in a linear search (n = 3266). Results were validated for 1% estimated FDR using the built-in validation algorithm as described in [8]. The number of identified crosslinks varies a lot more compared to validation with xiFDR [1]. The biggest difference is observed for replicate one where the filtered database search reports 81 more crosslinks, more than double the amount of the search with the proteome-wide database. Interestingly, the difference is comparably small for replicate three, where the filtered proteome search only amounts in 12 more crosslink identifications. Overall 276 unique crosslinks are identified across the three replicates using the filtered proteome and 201 unique crosslinks using the full proteome.

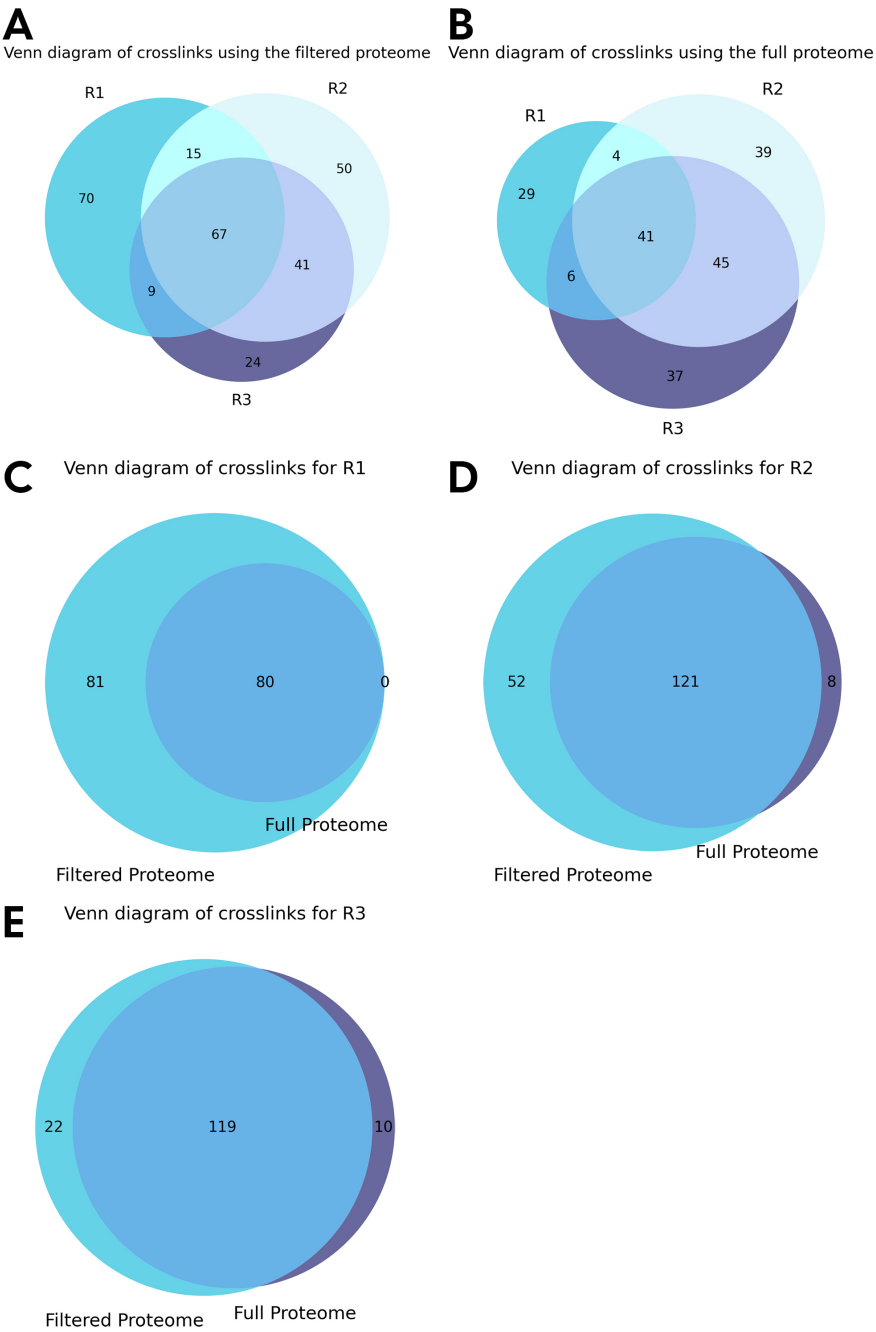


Fig. 3 Overlaps of identified crosslinks at 1% FDR in the *C. elegans* dataset using MS Annika for search and validation. A) Overlaps of identified crosslinks of the three biological replicates using the filtered proteome. B) Overlaps of identified crosslinks of the three biological replicates using the full proteome. C) Overlap of identified crosslinks for filtered and full proteome in replicate 1. D) Overlap of identified crosslinks for filtered and full proteome in replicate 2. E) Overlap of identified crosslinks for filtered and full proteome in replicate 3.

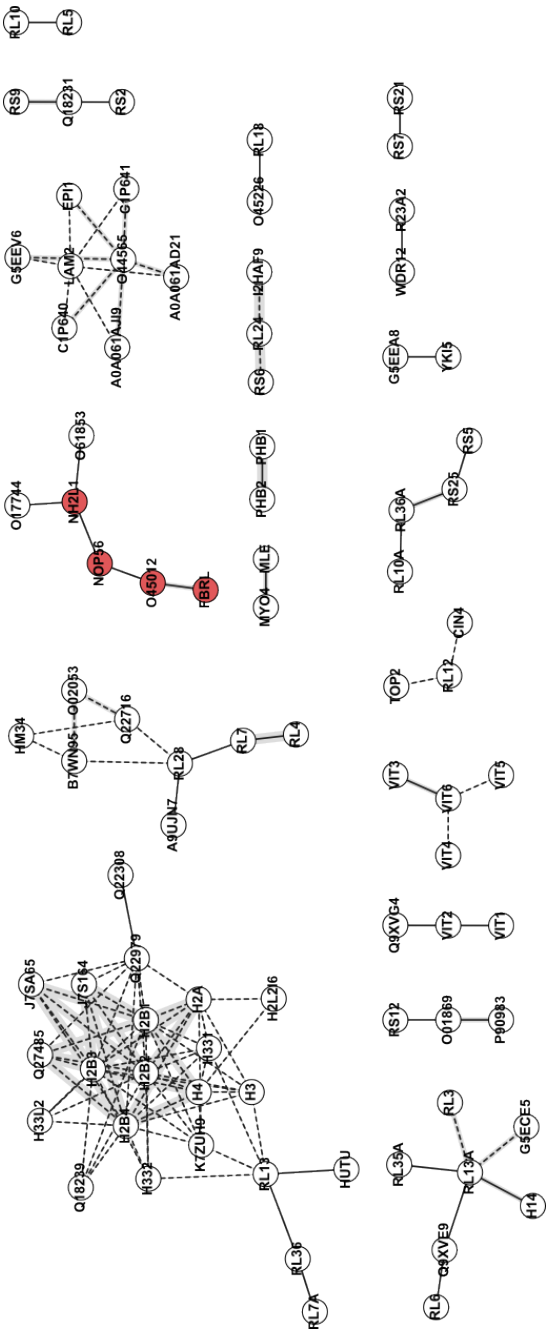


Fig. 4 Protein-protein interaction (PPI) network uncovered by the proteome-wide crosslink search of the *C. elegans* data and validated for 1% estimated PPI FDR using xiFDR [1]. Noticeably a large part of the identified inter crosslinks that satisfy 1% PPI FDR are ambiguous crosslinks connecting peptides that are shared among several proteins (depicted as dashed lines). Proteins labeled in red are part of the Box C/D RNP complex that we studied further in this manuscript. The network was created using xiView [9].

Table 1 Search times for crosslink identification in the three replicates of our *C. elegans* dataset using the full proteome-wide search.

Replicate	Nr. of Mass Spectra	Search Time
Replicate 1	556013	5h 51min
Replicate 2	576443	6h 28min
Replicate 3	515924	5h 40min

Table 2 Hardware configuration of the system that we used for benchmarking the different matrix multiplication approaches and for the more demanding proteome-wide searches.

Component	Product
MB	ASUS ROG Strix B650E-I
CPU	AMD Ryzen R9 7900X [12 cores @ 4.7 GHz base / 5.6 GHz boost]
Cooler	Corsair H115i RGB Elite [280mm]
Paste	Corsair
RAM	Kingston 64 GB DDR5 RAM [5600 MT/s, 36 CAS]
GPU	ASUS Dual [Nvidia] GeForce RTX 4060 Ti OC [16 GB VRAM]
SSD/HDD	Corsair MP600 Pro NH 2 TB NVMe SSD [PCIe 4.0]
OS	Windows 11 Pro 64-bit (Version 10.0, Build 22631)

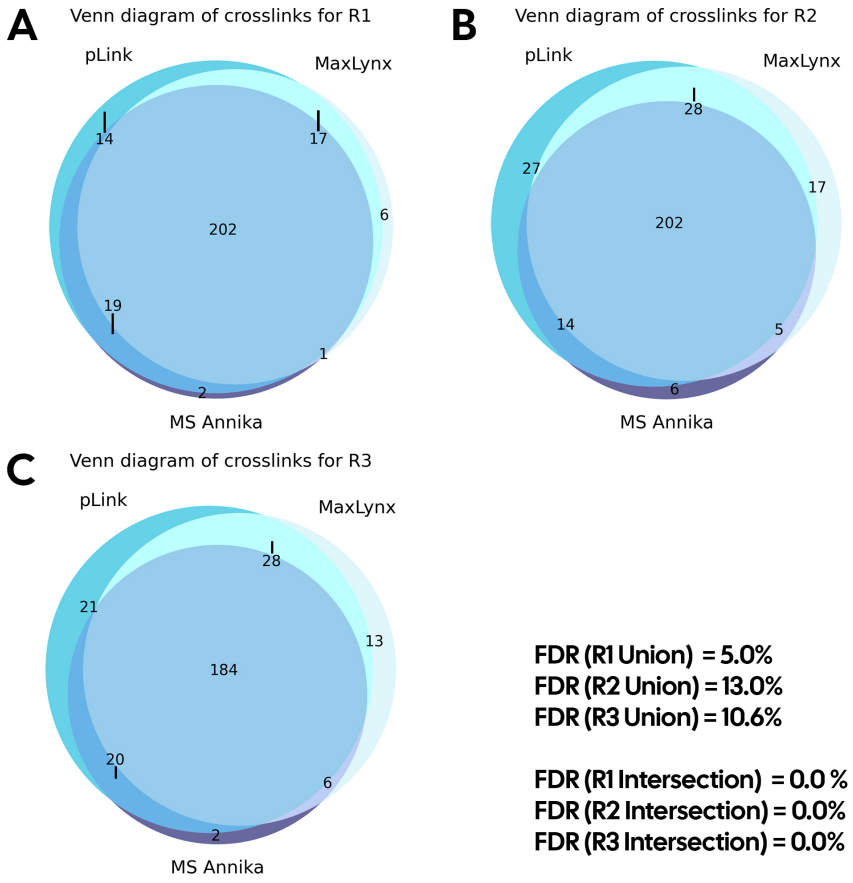


Fig. 5 Overlaps of identified crosslinks at 1% FDR in the dataset by Beveridge and co-workers [2]. All venn diagrams show overlaps of identifications from the search engines MaxLynx [10], pLink [11] and MS Annika. A) Overlaps of crosslink identifications for replicate 1. B) Overlaps of crosslink identifications for replicate 2. C) Overlaps of crosslink identifications for replicate 3. The bottom right lists experimentally validated FDRs for unions and intersections, where the number after the "R" denotes the replicate, union denotes taking all crosslinks from all search engines, and intersection denotes only taking crosslinks that are identified by all three search engines: as expected the unions show higher experimentally validated FDRs while intersections report zero false positive hits.

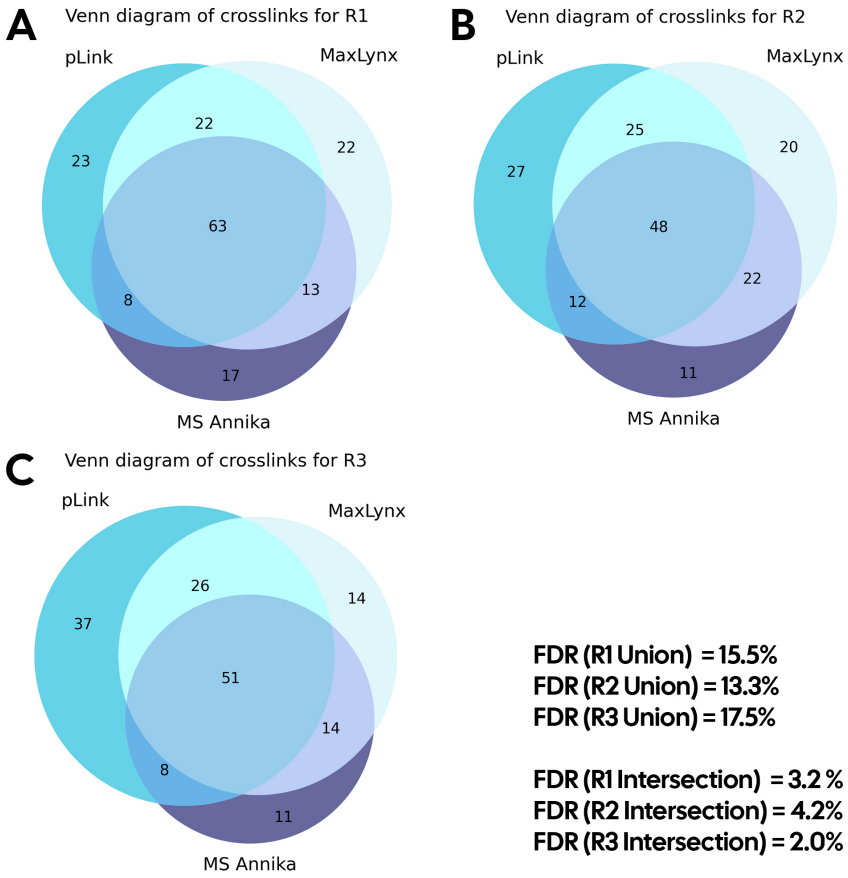


Fig. 6 Overlaps of identified crosslinks at 1% FDR in the dataset by Matzinger and co-workers [3]. All venn diagrams show overlaps of identifications from the search engines MaxLynx [10], pLink [11] and MS Annika. A) Overlaps of crosslink identifications for replicate 1. B) Overlaps of crosslink identifications for replicate 2. C) Overlaps of crosslink identifications for replicate 3. The bottom right lists experimentally validated FDRs for unions and intersections, where the number after the "R" denotes the replicate, union denotes taking all crosslinks from all search engines, and intersection denotes only taking crosslinks that are identified by all three search engines: remarkably intersections still consist of up to two false positive hits and do not meet the 1% FDR criteria.

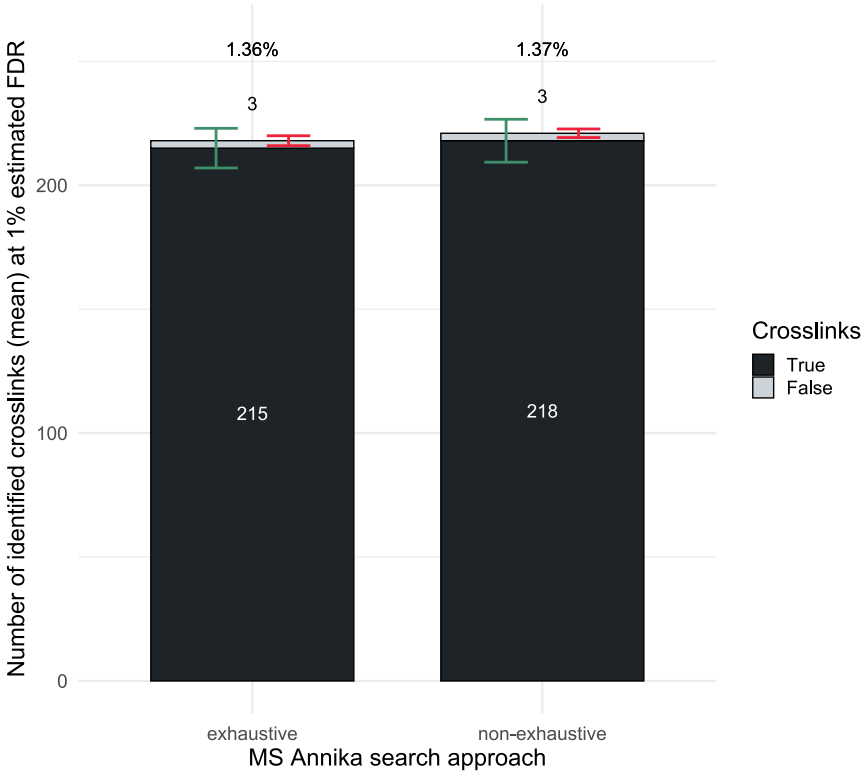


Fig. 7 Comparison of the MS Annika exhaustive and non-exhaustive search approach using the dataset by Beveridge and co-workers [2]. While the non-exhaustive approach comes with the inherent risk of potentially missing true positive crosslink identifications it still performs arguably better or at least on par in terms of identifications at 1% estimated FDR, yielding 218 true and 3 false crosslinks on average while the exhaustive search reports 215 true and 3 false crosslinks on average. All numbers are averages from three technical replicates (n = 3), the number of crosslinks was rounded to the closest integer value. Percentage numbers above the bars denote the average calculated experimentally validated FDR rounded to two decimal places. Error bars denote the standard deviation, in green for true positive crosslinks and in red for false positive crosslinks.

Table 3 Number of target and decoy hits for the exhaustive and non-exhaustive search approach of MS Annika using the dataset by Beveridge and co-workers [2].

Search Approach	Replicate	Target Hits	Decoy Hits
Exhaustive	1	423	343
Non-Exhaustive	1	265	35
Exhaustive	2	771	646
Non-Exhaustive	2	313	68
Exhaustive	3	746	581
Non-Exhaustive	3	289	65

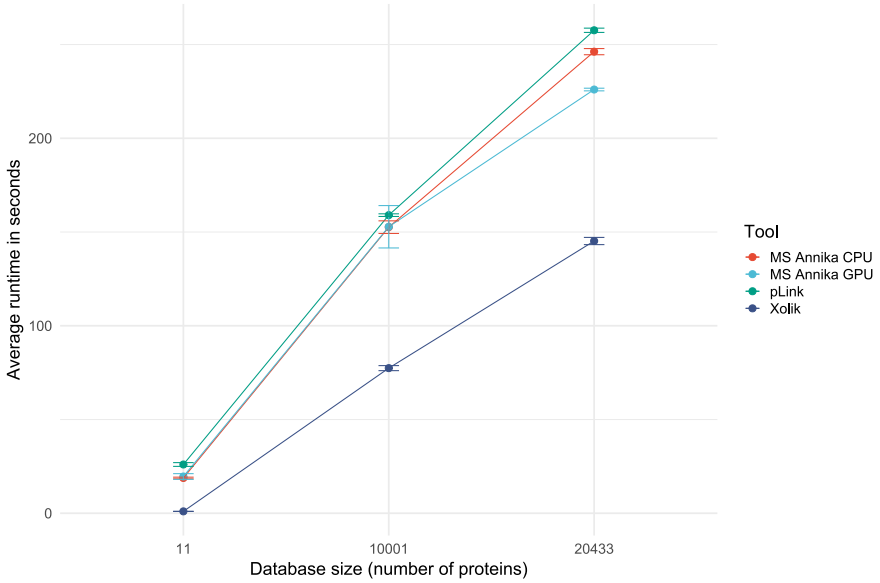


Fig. 8 Comparison of runtimes of the different crosslink search engines capable of processing large proteome-wide studies except Kojak [4, 5]. MS Annika, pLink [11] and Xolik [12] were used to search replicate one of the dataset by Beveridge and co-workers [2] employing protein databases of increasing size up to human proteome scale. Xolik outperforms all other tools in search speed, taking first place regardless of database size while MS Annika slightly outperforms pLink. All depicted values are averages from five consecutively repeated runs (n = 5). Error bars denote the standard deviation.

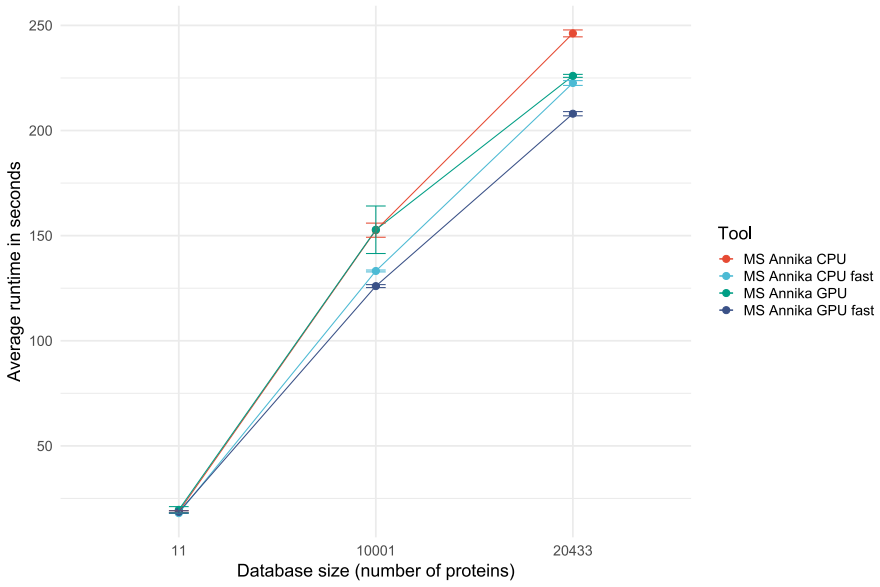


Fig. 9 Comparison of runtimes of the different MS Annika search approaches. Expectedly the "fast" GPU-based approach (see 4 performs the best with the "fast" CPU-based approach shortly behind. Generally the "fast" approach outperforms the standard approach and GPU-based outperforms CPU-based. All depicted values are averages from five consecutively repeated runs (n = 5). Error bars denote the standard deviation.

Table 4 Average runtimes of the different crosslink search engines using various protein database sizes. MS Annika was run with default parameters except for "fast" searches where parameter "Top N Filter" was reduced from 10 to 2.

Search Engine	Mean Runtime (s)	SD (s)	Database Size
MS Annika CPU	18,8	0,447213595	11
MS Annika CPU	152,6	3,361547263	10001
MS Annika CPU	246,2	1,643167673	20433
MS Annika CPU fast	18	0	11
MS Annika CPU fast	133,2	0,447213595	10001
MS Annika CPU fast	222,6	1,140175425	20433
MS Annika GPU	19,6	1,516575089	11
MS Annika GPU	152,8	11,30044247	10001
MS Annika GPU	226	0,707106781	20433
MS Annika GPU fast	18,6	0,547722558	11
MS Annika GPU fast	126	0,707106781	10001
MS Annika GPU fast	208	1	20433
pLink	26	1	11
pLink	159	0,707106781	10001
pLink	257,6	1,140175425	20433
Xolik	1	0	11
Xolik	77,4	1,341640786	10001
Xolik	145,2	1,923538406	20433
Kojak	8,8	0,447213595	11
Kojak	778,4	87,17109613	10001
Kojak	1229,2	24,84351022	20433

Table 5 Hardware configuration of the system that we used for runtime analysis of the different crosslink search engines.

Component	Product
MB	ASUS ROG Crosshair X670E Extreme
CPU	AMD Ryzen R9 7950X [16 cores @ 4.5 GHz base / 5.7 GHz boost]
Cooler	Corsair H150i Elite Capellix XT [360mm]
Paste	Thermal Grizzly Kryonaut
RAM	Corsair 64 GB DDR5 RAM [5600 MT/s, 30 CAS]
GPU	ASUS ROG Strix [Nvidia GeForce RTX] 4090 OC [24 GB VRAM]
SSD/HDD	Corsair MP600 Pro NH 8 TB NVMe SSD [PCIe 4.0]
OS	Windows 11 Pro 64-bit (Version 10.0, Build 22631)

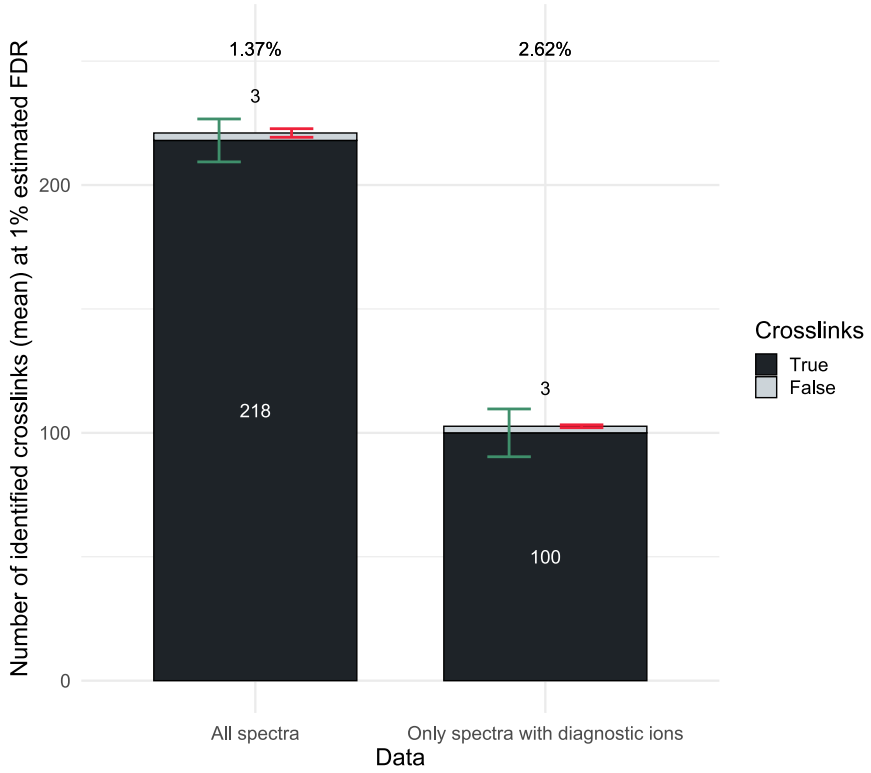


Fig. 10 Comparison of running the MS Annika non-cleavable search on either all spectra or only spectra containing diagnostic ions of the benchmark dataset by Beveridge and co-workers [2]. Excluding spectra without diagnostic ions results in a substantial loss of identified true positive crosslinks, yielding less than half of the crosslinks that are detected when all spectra are searched. Because the number of false positive hits remains the same, this also constitutes a higher experimentally validated FDR of 2.62% and therefore an overall worse result. Results were validated for 1% estimated FDR. All numbers are averages from three technical replicates (n = 3), the number of crosslinks was rounded to the closest integer value. Percentage numbers above the bars denote the average calculated experimentally validated FDR rounded to two decimal places. Error bars denote the standard deviation, in green for true positive crosslinks and in red for false positive crosslinks.

Table 6 Number of mass spectra containing diagnostic ions in the dataset by Beveridge and co-workers [2].

Replicate	Nr. of Mass Spectra	Nr. of Mass Spectra with Diagnostic Ions
Replicate 1	5198	883
Replicate 2	10538	2280
Replicate 3	9892	1966

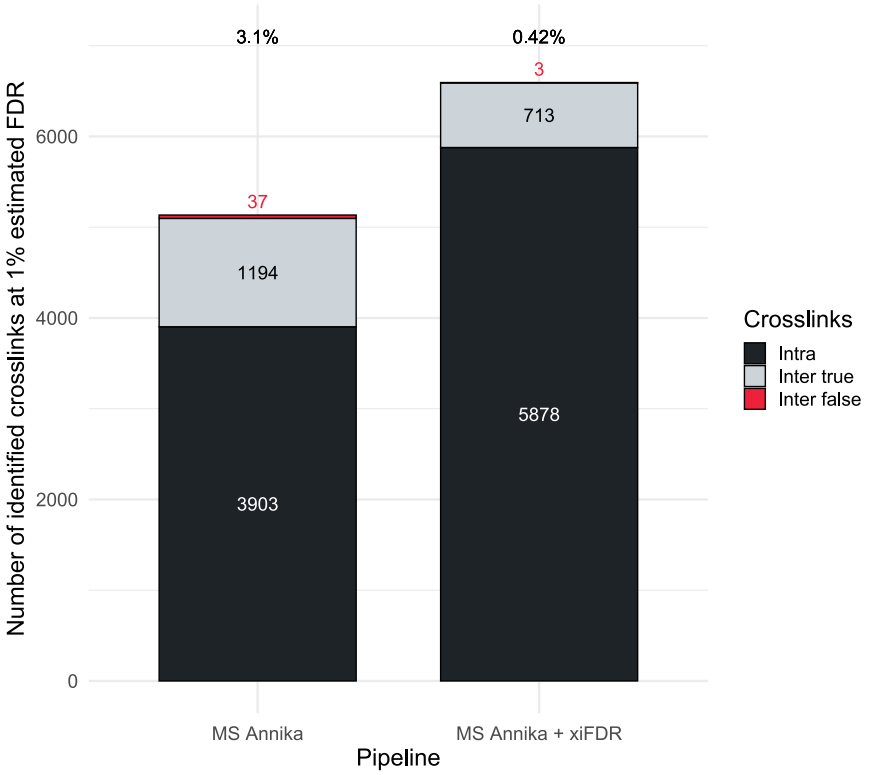


Fig. 11 Comparison of crosslink results of MS Annika using either the built-in FDR validation or xiFDR [1] for validation, as demonstrated on the dataset by Lenz and co-workers [6]. Using more sophisticated validation approaches like xiFDR reduces the experimentally validated inter crosslink FDR from 3.1% to 0.42% while overall boosting the number of crosslink identifications from 5134 to 6594. Results were validated for 1% estimated FDR. Percentage numbers above the bars denote the calculated experimentally validated inter crosslink FDR rounded to two decimal places

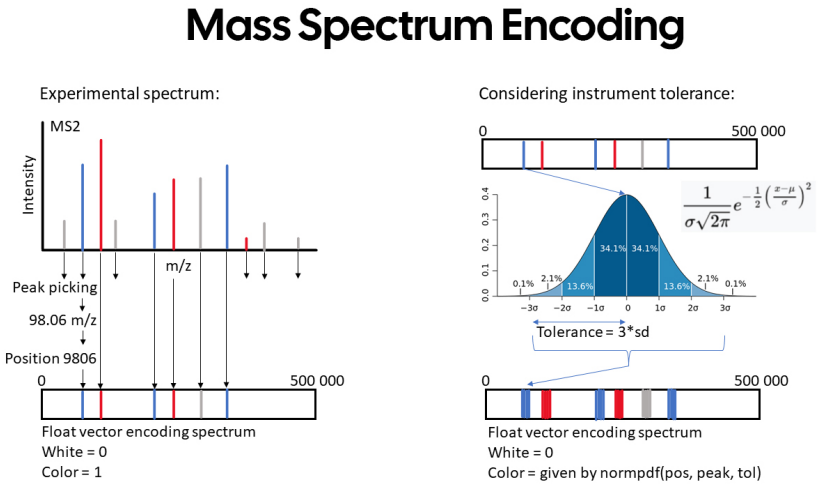


Fig. 12 Visualization of the mass spectrum encoding. For every mass spectrum a 500 000-dimensional float vector is created and all values are initialized to zero. Mass spectra may be peak picked or otherwise preprocessed before encoding to avoid the incorporation of noise, e.g. by using the IMP MS2 Spectrum Processor node or other deconvolution algorithms. Subsequently, for every peak the m/z is taken, multiplied by 100 and rounded to the closest integer. If the resulting value is smaller than 500 000, the element at the vector index equal to the value is set to one. To account for instrument tolerance every peak is modelled as normal distribution with μ equal to the peak's m/z and σ equal to tolerance/3, where tolerance is a user-definable parameter. The encoding works the same way, however instead of setting the elements in the vector to one, the resulting value of the normal distribution is taken, as shown on the right-hand side of the figure. Depiction of the normal distribution is taken from Wikimedia Commons, author: Ainali, licensed unter CC-BY-SA 3.0 https://commons.wikimedia.org/wiki/File:Standard_deviation_diagram_micro.svg.

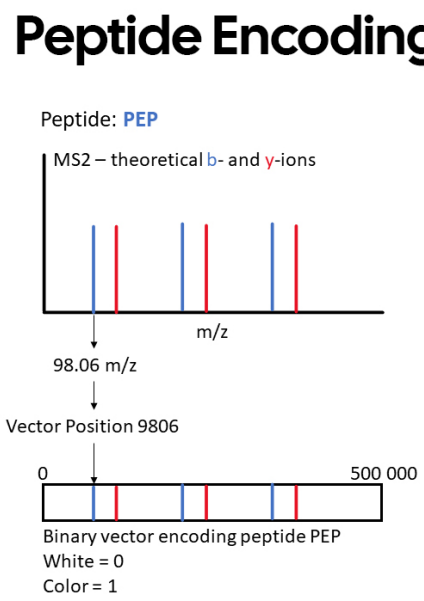


Fig. 13 Visualization of the peptide encoding. Analogous to the mass spectrum encoding, every peptide is encoded as a 500 000-dimensional float vector. For every peptide such a vector is created and all values are initialized to zero. Furthermore, after calculating all theoretical ions for the given peptide, every ion's m/z value is multiplied by 100 and rounded to the closest integer. If the resulting value is smaller than 500 000, the element at the vector's corresponding index is set to one.

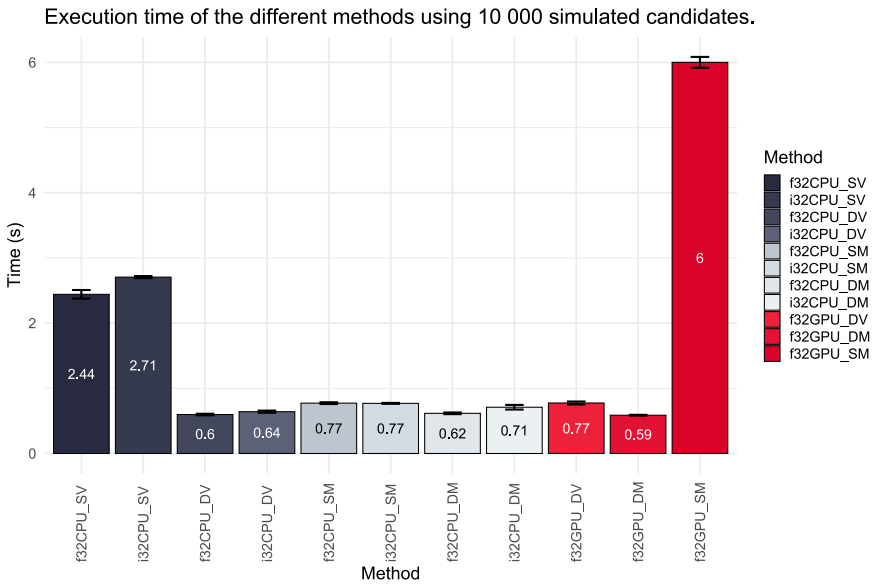


Fig. 14 Synthetic benchmark of the different sparse matrix multiplication algorithms implemented in MS Annika. The benchmark consisted of simulating 10 000 peptides with 100 theoretical ions each, as well as 1001 simulated mass spectra with 1000 peaks each. For this benchmark the f32-based sparse matrix * dense matrix multiplication algorithm on the GPU performed best at an average of 0.59 seconds. The specific algorithms are described in the methods section in the main text. Values are averages of five different runs, error bars denote the standard deviation (n = 5). More synthetic benchmarks are available at https://github.com/hgb-bin-proteomics/CandidateVectorSearch/blob/master/benchmarks.md.

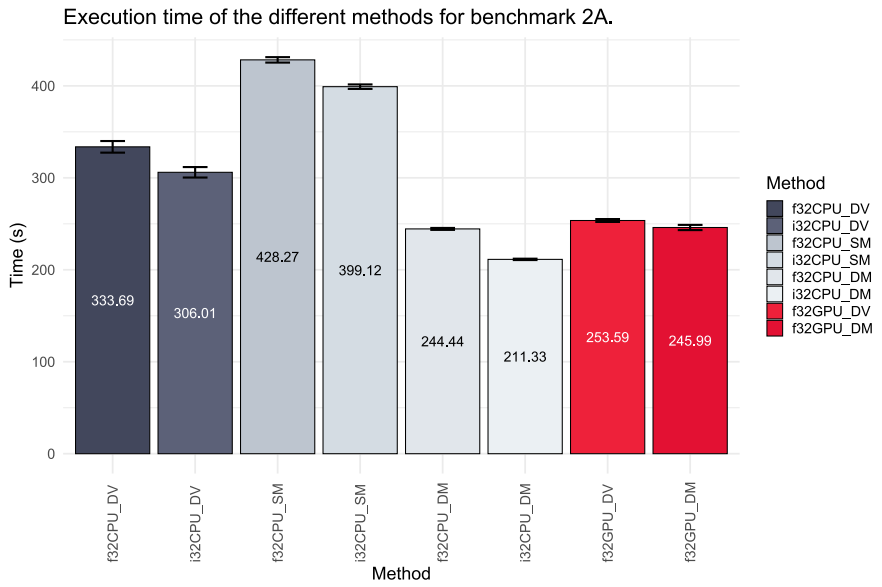


Fig. 15 Benchmark of the different sparse matrix multiplication algorithms implemented in MS Annika using real data. Mass spectra of replicate 1 of the dataset by Beveridge and co-workers [2] were searched against a protein database consisting of the sequence of *S. pyogenes* Cas9 and the human SwissProt proteome. Some algorithms were omitted from benchmarking due to their long runtimes which were already immanent from the synthetic benchmarks, specifically *f32CPU_SV*, *i32CPU_SV* and *f32GPU_SM*. The results show that integer-based algorithms generally outperform the float-based counterparts and matrix * matrix multiplication outperforms matrix * vector multiplication. For our system (see 2) i32-based sparse matrix * dense matrix multiplication performed the best at an average of 211.33 seconds. The specific algorithms are described in the methods section in the main text. Values are averages of five different runs, error bars denote the standard deviation (n = 5). More benchmarks are available at https://github.com/hgb-bin-proteomics/CandidateSearch/blob/master/benchmarks.md.

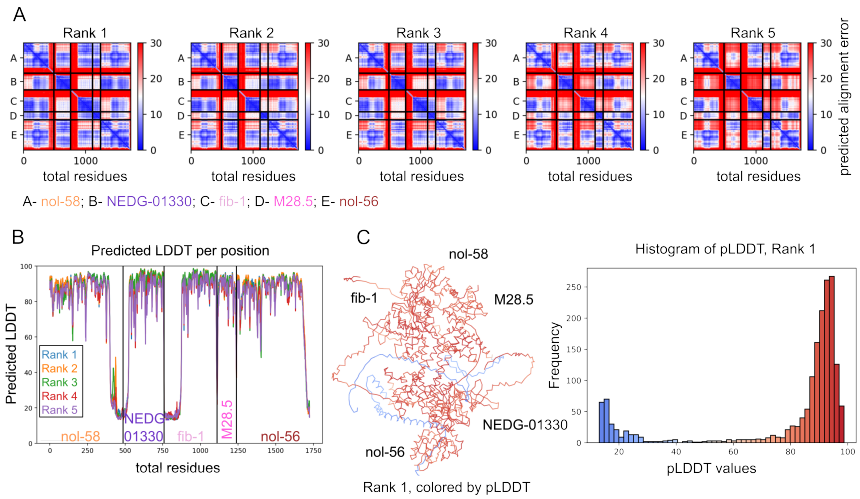


Fig. 16 Diagnostic plots for the Box C/D RNP complex prediction. The nol-58, nol-56, M28.5 and fib-1 are predicted with high confidence to bind distinct binding sites. The interface between NEDG-01330 and nol-56 is also predicted with high confidence, while the n-terminal region of nol-56 and nol-58 resulted in low prediction accuracy. Shown are A: the PAE plot (Rank 1-5), B: the pLDDT plot of each rank separately, C: the structure of the top-ranked model in Ca trace coloured by pLDDT, and D: A histogram plot showing the distribution of pLDDT values of the Rank 1 structure, coloured by the confidence of the prediction (red: high confidence, blue: low confidence).

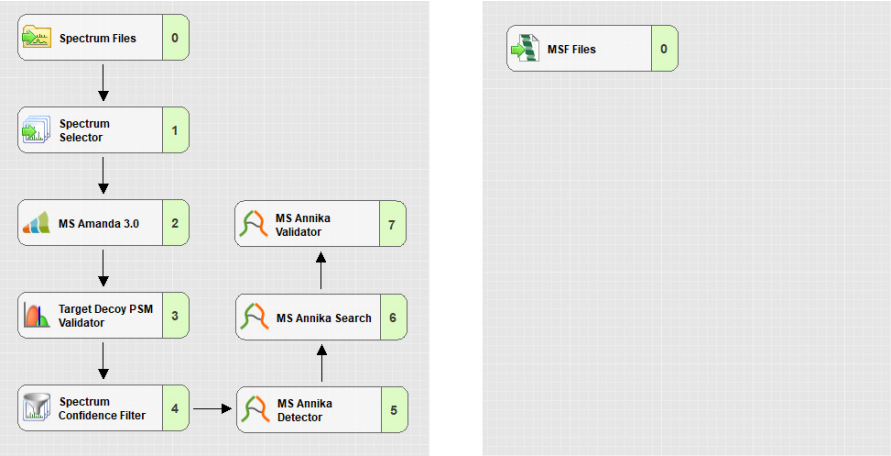


Fig. 17 The standard Proteome Discoverer workflow for crosslink identification with MS Annika. Linear and monolinked peptides are first identified with MS Amanda [13–15]. Mass spectra with associated high-confidence peptide spectrum matches (PSMs) are filtered out and not considered for crosslink search. Crosslinks are identified and validated with the MS Annika Detector node, the MS Annika Search node, and the MS Annika Validator node. The left panel shows the processing workflow while the right panel shows the consensus workflow.

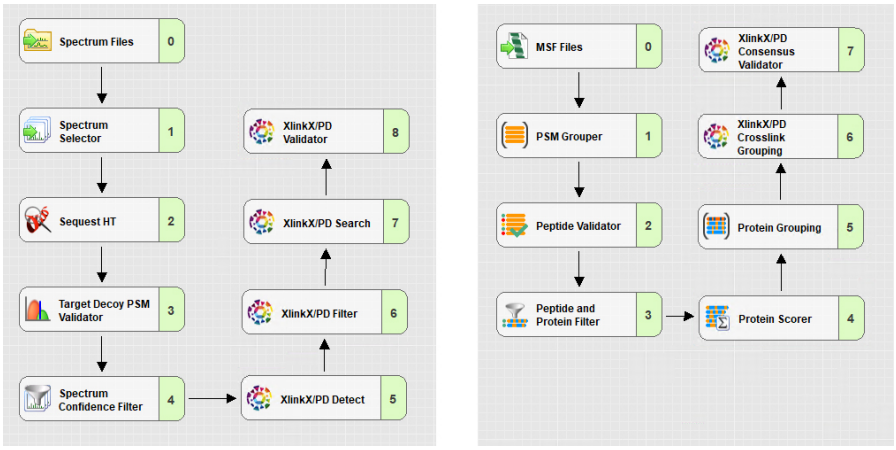


Fig. 18 The non-cleavable HCD/CID MS2 Proteome Discoverer workflow for crosslink identification with XlinkX [7]. Linear and monolinked peptides are first identified with the Sequest HT node [**?**]. Mass spectra with associated high-confidence peptide spectrum matches (PSMs) are filtered out and not considered for crosslink search. Crosslinks are identified and validated with the XlinkX nodes. The left panel shows the processing workflow while the right panel shows the consensus workflow.

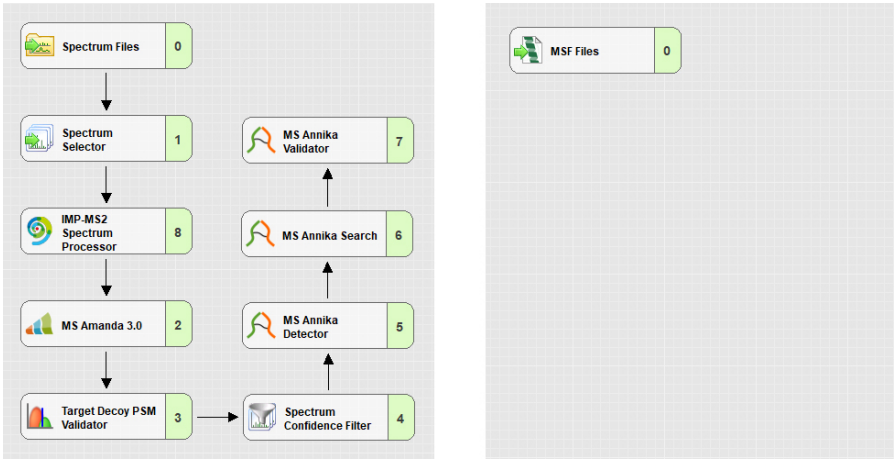


Fig. 19 The Proteome Discoverer workflow for crosslink identification with MS Annika for complex samples. Mass spectra are first deisotoped with the IMP MS2 Spectrum Processor node. Then linear and monolinked peptides are identified with MS Amanda [13–15] and mass spectra with associated high-confidence peptide spectrum matches (PSMs) are filtered out and not considered for crosslink search. Crosslinks are identified and validated with the MS Annika Detector node, the MS Annika Search node, and the MS Annika Validator node. The left panel shows the processing workflow while the right panel shows the consensus workflow.

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